

Darwin Del Castillo, MD, MPH

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EDUCATION

Master of Public Health (MPH)
UNIVERSITY OF WASHINGTON

Sep 2023 – Jun 2025
SEATTLE – USA

Doctor of Medicine (MD)
UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS

Mar 2014 – Dec 2020
LIMA – PERU

ADDITIONAL EDUCATION

Professional Certificate in Clinical Research
UNIVERSIDAD PERUANA CAYETANO HEREDIA

Sep 2021 – Apr 2022
LIMA – PERU

SKILLS

Programming: R, Python, SQL (MySQL), SAS

Cloud Computing: Slurm HPC, Linux (bash)

Tools: Quarto, Shiny (R), REDCap, Git (GitHub, GitLab)

Data Standards: ICH E6, HIPAA, GDPR/UK GDPR, ICD-10/ICD-9, SNOMED CT

Packages: tidyverse, survival, lme4, rstan, mice, cmprsk, ivreg, metafor

Statistical methods: Bayesian hierarchical modelling (MCMC), GLM/GLMM, time-to-event models, marginal structural models, propensity score methods, competing risks, multiple imputation, meta-analysis, uncertainty estimation

Languages: English (fluent), Spanish (native)

Core Competencies: Real-world evidence generation, observational study design, scientific writing, stakeholder communication

PROFESSIONAL EXPERIENCE

Research Assistant – MRC CSO Social and Public Health Sciences Unit
SCHOOL OF HEALTH AND WELLBEING – UNIVERSITY OF GLASGOW

Dec 2025 – Current
GLASGOW – UK

- Estimated the causal effects of unemployment on physical and mental health outcomes using g-methods with survey data.
- Calculated the causal direct and indirect effects of relative income on all-cause mortality using linked administrative data and g-methods.
- Modelled the effect of implementing economic policies on health outcomes using microsimulation modelling.

Research Assistant – Clinical Informatics
INSTITUTE FOR HEALTH METRICS AND EVALUATION

Jul 2024 – Jun 2025
SEATTLE – USA

- Developed hyperparameters for a Bayesian hierarchical meta-regression model to estimate prevalence, incidence, and clinical-outcome occurrence of lower respiratory tract infections across 225 countries.
- Modelled using data-driven priors and an MCMC algorithm the distribution of inclusion health populations in 22 high-income countries.
- Produced a global map of study locations using GBD 2023 georeference data to visualize geographic coverage and data gaps, informing meta-analysis data sources.

- Harmonized claims and inpatient data from 20+ hospitals (>5 million encounters) using ICD-10 phenotyping and probabilistic linkage.
- Built regression and survival analysis pipelines in R with quality control for reimbursement policy analysis.
- Conducted budget impact and cost-effectiveness analyses with uncertainty quantification for decision-makers.

Epidemiology Trainee
CRONICAS CENTER OF EXCELLENCE IN CHRONIC DISEASES

Mar 2019 – Aug 2023
LIMA – PERU

- Analyzed longitudinal data from population cohorts and RCTs using generalised linear methods and survival curves in cardiometabolic disease epidemiology.
- Conducted preliminary data analysis on clinical, demographic, and biological biomarkers to support research grant proposals.
- Co-authored 5 scientific manuscripts, which included longitudinal data analyses and machine learning models.

LICENSURE AND CERTIFICATION

National Registry of Investigators, RENACYT-CONCYTEC, Investigator Level VII (P0160977)	2024
CITI Program, Good Clinical Practice (ICH E6), Credential ID 65079238	2023
CITI Program, Human Subjects Learners, Credential ID 65079240	2023
CITI Program, Biomedical Responsible Conduct of Research, Credential ID 65079239	2023
Peruvian Medical Board, License No. 94534	2021

PUBLICATIONS ORCID: 0000-0002-8609-0312

Peer-Reviewed Publications

1. **Del Castillo-Fernández D**, Brañez-Condorena A, Villacorta-Landeo P, Saavedra-Garcia L, Bernabé-Ortiz A, Miranda J. Advances in the investigation of chronic non-communicable diseases in Peru. *An Fac med.* 2021;81(4) doi:10.15381/anales.v81i4.18798.
2. Lapidaire W, Proaño A, Blumenberg C, Loret de Mola C, Delgado CA, **Del Castillo D**, et al. Effect of preterm birth on growth and blood pressure in adulthood in the Pelotas 1993 cohort. *Int. J. Epidemiol.* 2023;52(6):1870–1877 doi:10.1093/ije/dyad084.
3. Vidyasagaran AL, Ayesha R, Boehnke JR, Kirkham J, Rose L, Hurst JR, et al. Core outcome sets for trials of interventions to prevent and to treat multimorbidity in adults in low and middle-income countries: the {COSMOS} study. *BMJ Glob. Health.* 2024;9(8):e015120 doi:10.1136/bmjgh-2024-015120.
4. Bazo-Alvarez JC, **Del Castillo D**, Piza L, Bernabé-Ortiz A, Carrillo-Larco RM, Smeeth L, et al. Multimorbidity patterns, sociodemographic characteristics, and mortality: Data science insights from low-resource settings. *Am. J. Epidemiol.* 2026;194(4):901–911 doi:10.1093/aje/kwae466.
5. Pinzon Cortes JA, Narongkiatikhun P, Fuhri Snethlage CM, **Del Castillo D**, Tommerdahl KL, Nadeau KJ, et al. Beyond glycemia in {T1D}: The hidden burden of insulin resistance and cardiorenal risk in type 1 diabetes - Part 1. *Endocrinol. Metab. Clin. North Am.* 2026 doi:10.1016/j.ecl.2026.04.008.
6. Pinzon Cortes JA, Narongkiatikhun P, Snethlage CF, **Del Castillo D**, Tommerdahl KL, Nadeau KJ, et al. Beyond glycemia in {T1D}: The hidden burden of insulin resistance and cardiorenal risk in type 1 diabetes - Part 2. *Endocrinol. Metab. Clin. North Am.* 2026 doi:10.1016/j.ecl.2026.04.014.